

SentixAI Platform — Complete Data Flow Diagram (DFD) Document

Version 4.2 Enterprise | BCA Semester 5, Parul University

Analyzing Unstructured Movie Review Feedback with Multi-Model AI

Project: SentixAI Platform (Version 4.2 Enterprise) Domain: Artificial Intelligence, NLP, Multi-Model Machine Learning Submitted By:

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1. Introduction to DFD

A **Data Flow Diagram (DFD)** is a structured analysis tool that graphically represents the flow of data through an information system. It visualizes how data enters the system, how it is processed and transformed, where it is stored, and how it exits.

Purpose of this document:

This DFD document provides a complete, multi-level decomposition of the **SentixAI Platform**, covering every subsystem from user authentication to AI-powered sentiment analysis and executive dashboard rendering. Each level progressively reveals more granular detail about the system's internal data transformations.

Leveling Strategy:

Level	Scope	Description
Level 0	Context Diagram	Shows SentixAI as a single process with all external entities
Level 1	Major Subsystems	Decomposes SentixAI into 5 core subsystems
Level 2.x	Detailed Decomposition	Each Level 1 subsystem is further broken into granular processes

2. Notation and Symbols Used

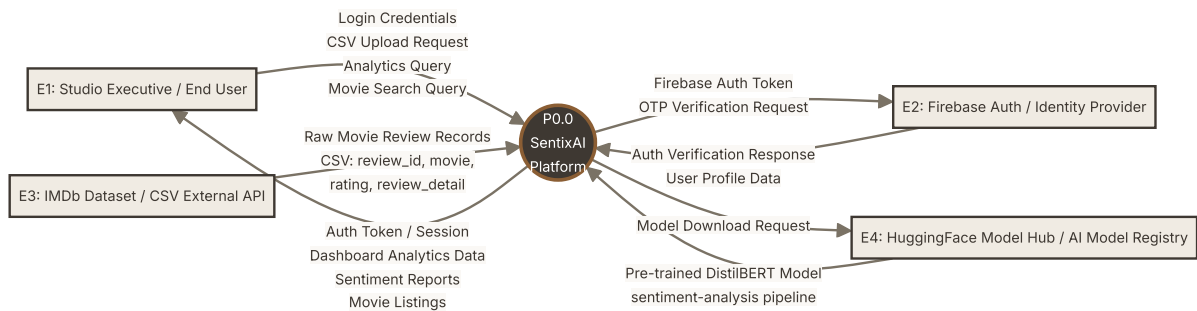
Symbol	Meaning	Example
Rectangle	External Entity (source/sink of data)	Studio Executive, IMDb API
Rounded Rectangle / Circle	Process (transforms data)	P1.0 — Authenticate User
Open Rectangle (parallel lines)	Data Store	D1 — User Database
Arrow	Data Flow (with label)	Login Credentials

Naming Convention:

- **Pn.m** — Process number (Level.Sub-process)
- **Dn** — Data Store number
- **En** — External Entity number

3. Level 0 — Context Diagram

The Context Diagram shows the **SentixAI Platform** as a single, monolithic process surrounded by all external entities that interact with it. This is the highest level of abstraction.

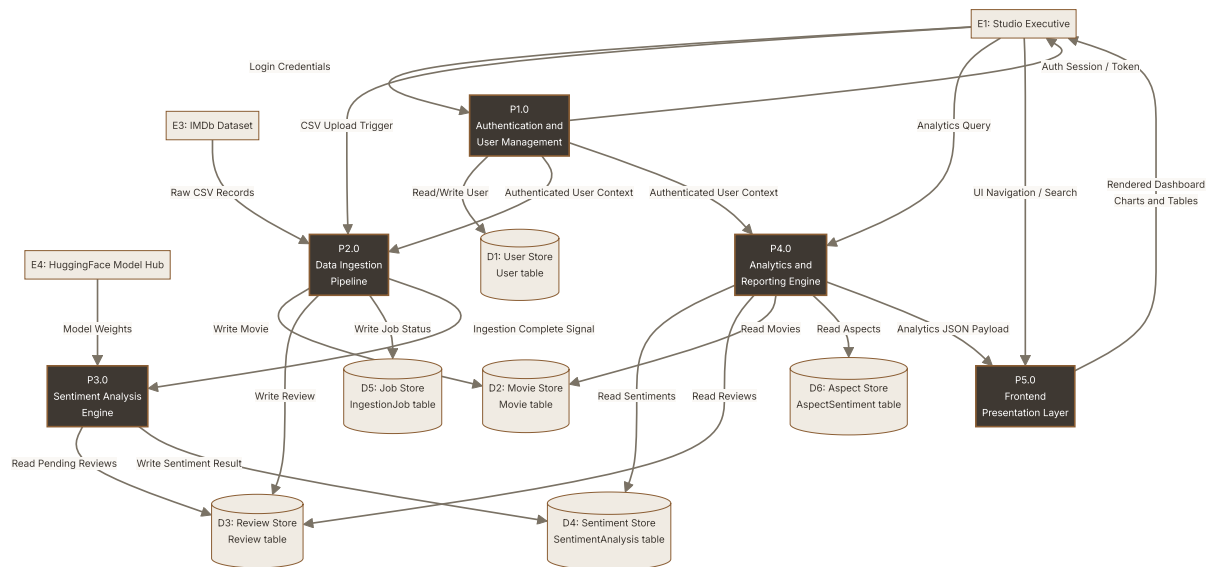


Level 0 — Data Flow Summary Table

No.	Data Flow	From	To	Description
DF-01	Login Credentials	E1: Studio Executive	P0: SentixAI	Email + Password / OTP for authentication
DF-02	Auth Token / Session	P0: SentixAI	E1: Studio Executive	JWT token or session for authorized access
DF-03	CSV Upload Request	E1: Studio Executive	P0: SentixAI	Trigger to ingest IMDb dataset with maxRecords
DF-04	Raw Movie Review Records	E3: IMDb Dataset	P0: SentixAI	CSV rows: review_id, movie, rating, review_detail, review_date
DF-05	Dashboard Analytics Data	P0: SentixAI	E1: Studio Executive	Sentiment overview, timeline charts, anomaly alerts
DF-06	Sentiment Reports	P0: SentixAI	E1: Studio Executive	Per-movie sentiment breakdown, aspect scores
DF-07	Pre-trained DistilBERT Model	E4: HuggingFace	P0: SentixAI	ML model weights for local inference
DF-08	Firebase Auth Response	E2: Firebase	P0: SentixAI	User identity verification result

4. Level 1 — Major Subsystem Decomposition

Level 1 decomposes the monolithic SentixAI Platform process into **5 major subsystems** with their interconnecting data flows and shared data stores.

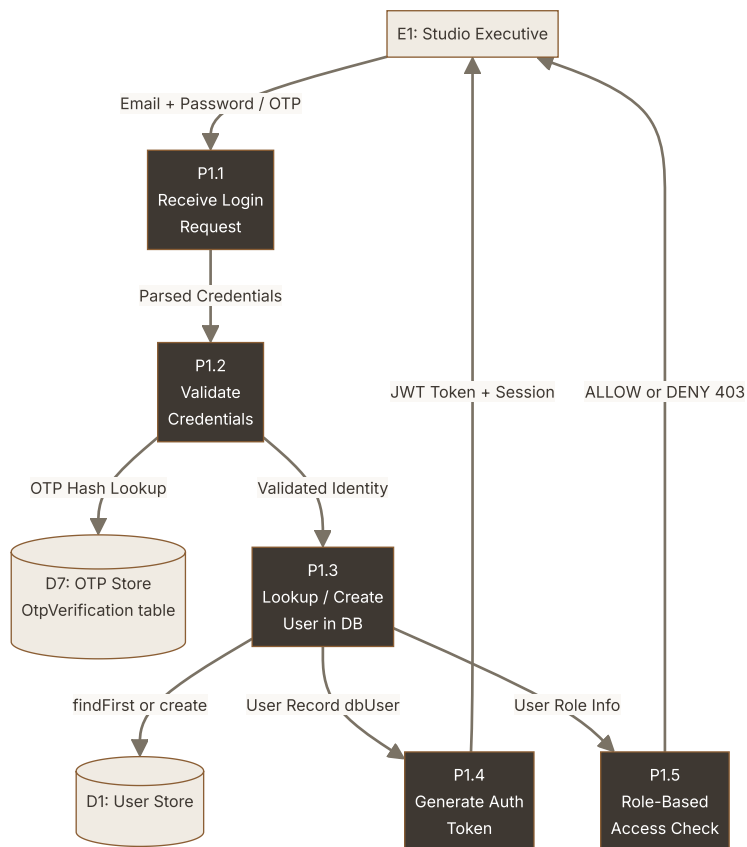


Level 1 — Process Summary

Process ID	Process Name	Input Data	Output Data	Data Stores Used
P1.0	Authentication and User Mgmt	Login Credentials, OTP	Auth Token, User Context	D1 (User)
P2.0	Data Ingestion Pipeline	Raw CSV Records, Upload Trigger	Normalized Reviews, Movies	D2, D3, D5
P3.0	Sentiment Analysis Engine	Pending Reviews, Model Weights	Sentiment Scores	D3, D4
P4.0	Analytics and Reporting Engine	Sentiments, Reviews, Movies	Analytics JSON	D2, D3, D4, D6
P5.0	Frontend Presentation Layer	Analytics JSON, User Context	Rendered UI (Charts, Tables)	Client-side only

5. Level 2.1 — Authentication and User Management

Decomposes P1.0 into its internal sub-processes.

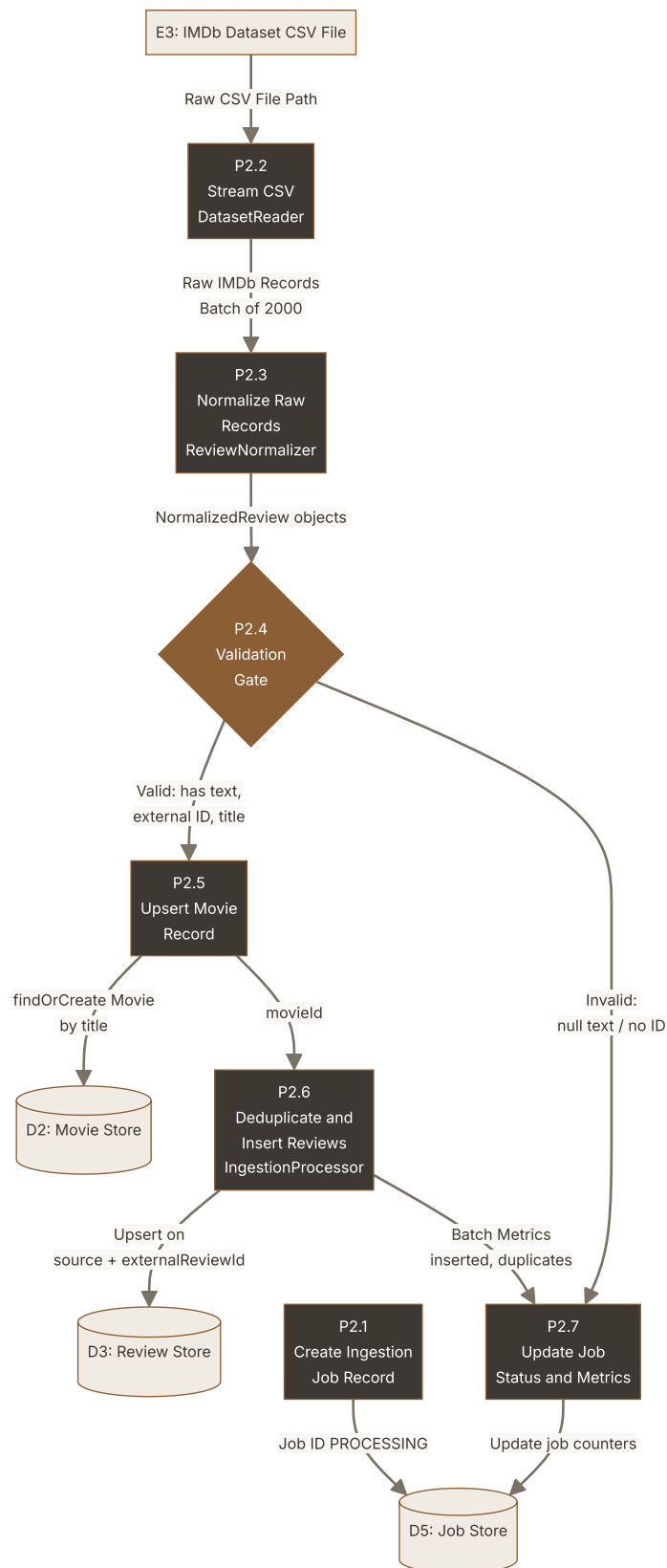


Level 2.1 — Sub-Process Detail

Process	Description	Implementation File
P1.1	Receives raw login request from Express router	user.routes.ts
P1.2	Validates OTP hash against stored hash, checks expiry and attempt count	auth.ts, OtpVerification model
P1.3	Queries User table via Prisma findFirst, auto-creates mock user in dev mode if none exists	auth.ts requireAuth middleware
P1.4	Issues JWT/session token for subsequent API calls via localStorage	Frontend AuthContext
P1.5	requireRole middleware checks dbUser.role against allowed roles USER or ADMIN	auth.ts

6. Level 2.2 — Data Ingestion Pipeline

Decomposes P2.0 — the 7-stage movie review processing workflow.



Level 2.2 — Sub-Process Detail

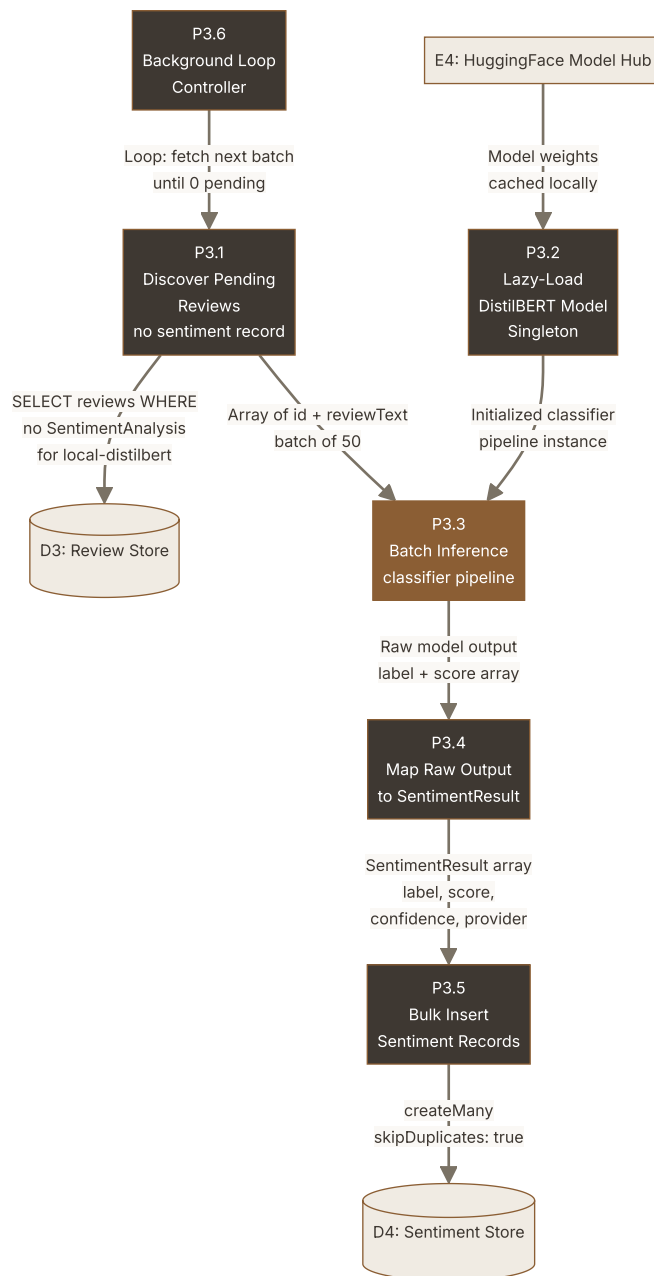
Process	Description	Key Logic	Implementation File
P2.1	Creates an IngestionJob record with status PROCESSING	db.ingestionJob.create	DatasetService.ts
P2.2	Streams CSV file in batches of 2,000 records using Node.js streams	csv-parse streaming parser	DatasetReader.ts
P2.3	Normalizes raw IMDb fields: strips HTML br tags, parses ratings 1-10, converts dates, resolves spoiler booleans	ReviewNormalizer.normalize	ReviewNormalizer.ts
P2.4	Validation gate: discards records with empty text, missing review_id, or missing movie title	Returns null for invalid	ReviewNormalizer.ts
P2.5	Finds or creates a Movie record using the movie title, assigns UUID	findFirst / create	IngestionProcessor.ts
P2.6	Deduplicates reviews using composite unique key source+externalReviewId, inserts new reviews via createMany	skipDuplicates: true	IngestionProcessor.ts
P2.7	Updates IngestionJob counters in real-time so UI can poll progress, sets final status to COMPLETED or FAILED	db.ingestionJob.update	DatasetService.ts

Data Transformation: Raw CSV to Normalized Review

Raw CSV Field	Transformation	Normalized Field
review_id: "rw1234567"	Trim whitespace	externalId: "rw1234567"
movie: "Dune: Part Two"	Trim whitespace	movieTitle: "Dune: Part Two"
reviewer: "john_doe"	Trim, nullable	reviewer: "john_doe"
rating: "9"	Parse to Number, validate 1-10	rating: 9
review_date: "3 May 2024"	Date.parse to ISO DateTime	reviewDate: 2024-05-03T00:00:00
spoiler_tag: "1"	Convert to Boolean	spoiler: true
review_detail: with br tags	Strip HTML, trim	text: cleaned text
helpful: "[15, 20]"	String conversion	helpfulVotes: "[15, 20]"
(none)	Auto-set	source: "IMDb Dataset"

7. Level 2.3 — Sentiment Analysis Engine

Decomposes P3.0 — the AI/ML core of SentixAI.



Level 2.3 — Sub-Process Detail

Process	Description	Implementation
P3.1	Queries Review table for records that have no matching SentimentAnalysis row where modelProvider equals local-distilbert	SentimentService.processPendingReviews
P3.2	Singleton lazy-loader for the DistilBERT model. Downloads from HuggingFace on first call, then caches locally using xenova/transformers	TransformersProvider.loadModel
P3.3	Runs batch inference using the sentiment-analysis pipeline. Processes array of review texts in parallel	TransformersProvider.analyzeBatch
P3.4	Maps raw transformer output label POSITIVE score 0.95 to normalized SentimentResult with sign-adjusted score. NEGATIVE becomes -0.95	TransformersProvider.mapToResult
P3.5	Bulk inserts all sentiment results into SentimentAnalysis table. Uses composite unique key reviewId + modelProvider to prevent duplicates	SentimentService.processPendingReviews
P3.6	Background async loop: repeatedly fetches batches of 50 pending reviews, processes them, yields event loop with 100ms delay, stops when 0 pending	SentimentService.startBackgroundProcessing

AI Model Pipeline Data Transformation

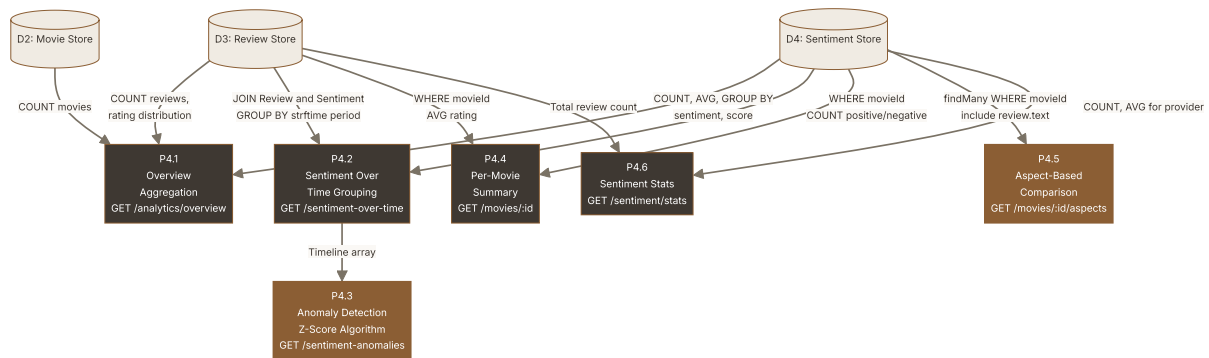
Stage	Input	Output
Text Input	"The cinematography was breathtaking but the plot dragged badly"	Raw text string
Model Inference	Raw text	label: POSITIVE, score: 0.9876
Score Mapping	Raw output	label: POSITIVE, score: +0.9876, confidence: 0.9876, provider: local-distilbert

Score Sign Convention:

- POSITIVE: score = +confidence (e.g., +0.98)
- NEGATIVE: score = -confidence (e.g., -0.92)

8. Level 2.4 — Analytics and Reporting Engine

Decomposes **P4.0** — the intelligence layer.



Level 2.4 — Analytics API Endpoint Map

Process	API Endpoint	Input	Output	Algorithm
P4.1	GET /api/analytics/overview	Auth token	totalReviews, analyzedReviews, positivePercentage, sentimentDistribution, ratingDistribution	Prisma count, aggregate, groupBy
P4.2	GET /api/analytics/sentiment-over-time	groupBy=month	Array of period, totalReviews, positive, negative, averageScore	Raw SQL: strtime + GROUP BY period
P4.3	GET /api/analytics/sentiment-anomalies	groupBy=month, threshold=2	Array of period, sentimentScore, expectedScore, deviation, severity	Statistical Z-Score: $z = (x - \text{mean}) / \text{stddev}$
P4.4	GET /api/analytics/movies/:movield	movield param	movie, reviewCount, positiveCount, negativeCount, positivePercentage	Prisma count with WHERE filter
P4.5	GET /api/analytics/movies/:id/aspects	movield param	Array of aspect, mentions, positive, negative, averageScore	Keyword-based aspect attribution across 8 categories
P4.6	GET /api/sentiment/stats	Auth token	totalReviews, analyzedReviews, pendingReviews, progressPercentage, isProcessing	Prisma count and aggregate

Anomaly Detection Algorithm (P4.3)

```

INPUT: timeline[] = [{period, averageScore}, ...]
OUTPUT: anomalies[] = [{period, sentimentScore, expectedScore, deviation, severity}]

1. Calculate Mean:      mean = SUM(averageScore) / N
2. Calculate Std Dev:   stddev = SQRT(SUM((score - mean)^2) / N)
3. For each data point:
  a. z-score = (score - mean) / stddev
  b. IF |z-score| >= threshold (default 2.0):

      Flag as ANOMALY

      severity = z > 0 ? HIGH_POSITIVE : HIGH_NEGATIVE

1. Return only flagged anomalies

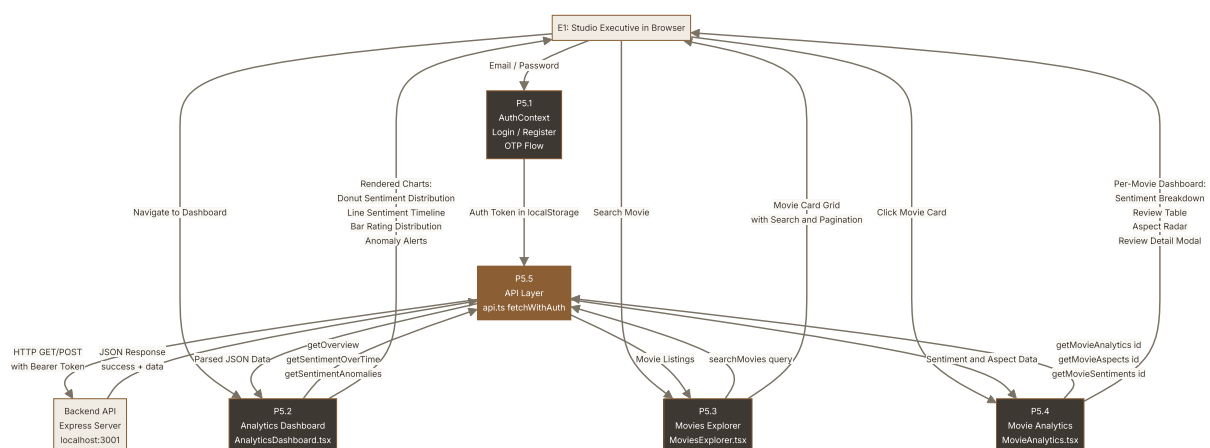
```

Aspect-Based Sentiment Analysis Keywords (P4.5)

Aspect	Detection Keywords
Acting	acting, actor, actress, cast, performance, played by
Story	story, plot, script, writing
Direction	director, direction, directed
Visuals	visuals, cinematography, camera, shot, lighting
Music	music, soundtrack, score, song, audio
Characters	character, protagonist, villain
Pacing	pacing, pace, slow, fast, dragged
Effects	effects, cgi, vfx, special effects

9. Level 2.5 — Frontend Presentation Layer

Decomposes P5.0 — the React/Vite client application.



Level 2.5 — Frontend Component Map

Component	File	API Calls	UI Elements
P5.1 AuthContext	contexts/AuthContext.tsx	/api/users/register, /api/users/login	Login form, OTP input, Register form
P5.2 Analytics Dashboard	components/AnalyticsDashboard.tsx	getOverview, getSentimentOverTime, getSentimentAnomalies	Donut chart, line chart, bar chart, anomaly alerts
P5.3 Movies Explorer	components/MoviesExplorer.tsx	searchMovies query	Search bar, movie card grid, pagination
P5.4 Movie Analytics	components/MovieAnalytics.tsx	getMovieAnalytics id, getMovieAspects id, getMovieSentiments id	Sentiment stats, aspect bars, review table, modal
P5.5 API Layer	api.ts	All HTTP calls with fetchWithAuth	N/A - data transport layer

10. Data Dictionary

Complete data dictionary for all data elements flowing through SentixAI.

Data Element	Type	Format	Constraints	Used In
userId	String	UUID v4	Primary Key, unique	D1
email	String	name@domain.com	Unique, required	D1, DF-01
firebaseUid	String	Alphanumeric	Unique, required	D1
role	String	USER or ADMIN	Default: USER	D1
status	String	ACTIVE or SUSPENDED	Default: ACTIVE	D1
movieId	String	UUID v4	Primary Key	D2
imdbId	String	tt + digits	Unique, optional	D2
title	String	Free text	Required, indexed	D2
reviewId	String	UUID v4	Primary Key	D3
externalReviewId	String	rw + digits	Composite unique with source	D3
reviewText	String	Free text	Required, non-empty	D3
rating	Integer	1 to 10	Nullable	D3
reviewDate	DateTime	ISO 8601	Nullable	D3
source	String	IMDB or METACRITIC	Default: IMDB	D3
sentiment	String	POSITIVE or NEGATIVE or NEUTRAL	Required	D4
score	Float	-1.0 to +1.0	Required	D4
confidence	Float	0.0 to 1.0	Nullable	D4
modelProvider	String	local-distilbert	Required	D4
jobStatus	String	PENDING or PROCESSING or COMPLETED or FAILED	Default: PENDING	D5
processedRecords	Integer	0 to N	Default: 0	D5
insertedReviews	Integer	0 to N	Default: 0	D5
otpHash	String	bcrypt hash	Required	D7
attempts	Integer	0 to 5	Default: 0, max 5	D7

11. Data Store Descriptions

Store ID	Store Name	Database Table	Description	Primary Key	Key Relationships
D1	User Store	User	Stores registered platform users with Firebase UID mapping	id UUID	Referenced by auth middleware
D2	Movie Store	Movie	Stores unique movie records with IMDb ID and metadata	id UUID	1:N to Review, 1:N to IngestionJob
D3	Review Store	Review	Stores normalized movie reviews from all sources	id UUID	N:1 to Movie, 1:N to SentimentAnalysis
D4	Sentiment Store	SentimentAnalysis	Stores AI-generated sentiment results per review per model	id UUID	N:1 to Review. Unique: reviewId + modelProvider
D5	Job Store	IngestionJob	Tracks dataset ingestion job progress and status	id UUID	N:1 to Movie optional
D6	Aspect Store	AspectSentiment	Stores aspect-level sentiment scores per review	id UUID	N:1 to Review, N:1 to Aspect
D7	OTP Store	OtpVerification	Temporary OTP verification codes for email auth	id UUID	Unique: email

12. External Entity Descriptions

Entity ID	Entity Name	Type	Description	Data Exchanged
E1	Studio Executive / End User	Human Actor	The primary user who uploads datasets, views dashboards, and analyzes movie sentiment	Login credentials, CSV upload triggers, analytics queries, rendered UI
E2	Firebase Authentication	External Service	Google identity platform providing OTP-based and email/password authentication	Auth tokens, user profile data, verification responses
E3	IMDb Dataset	External Data Source	CSV dataset containing 50,000+ raw movie reviews from IMDb with fields: review_id, movie, rating, review_detail	Raw CSV records
E4	HuggingFace Model Hub	External Service	Model registry hosting pre-trained transformer models. SentixAI uses Xenova/distilbert-base-uncased-finetuned-sst-2-english	Pre-trained model weights in ONNX format

13. Process Descriptions Mini-Spec

P1.0 — Authentication and User Management

Purpose: Validates user identity and establishes authorized session context for all API requests. **Trigger:** HTTP request to any protected endpoint via requireAuth middleware. **Logic:**

1. Extract Bearer token from Authorization header
2. Look up first user in database via db.user.findFirst
3. If no user exists: auto-create mock user for development
4. If user status is INACTIVE: return 403
5. Attach dbUser to Express req object and call next

P2.0 — Data Ingestion Pipeline

Purpose: Ingests, normalizes, validates, deduplicates, and stores raw movie reviews from CSV datasets into the relational database. **Trigger:** POST /api/ingestion/imdb with maxRecords parameter. **Logic:**

1. Create IngestionJob record with status PROCESSING
2. Open CSV file stream and process in batches of 2,000
3. For each batch: normalize fields, validate, find/create Movie, deduplicate, bulk insert Reviews
4. Update job counters after each batch for real-time progress tracking
5. On completion: set status COMPLETED. On error: set status FAILED with error message

P3.0 — Sentiment Analysis Engine

Purpose: Runs AI-powered sentiment classification on all unanalyzed reviews using a local DistilBERT transformer model. **Trigger:** POST /api/sentiment/analyze or background processing loop. **Logic:**

1. Query reviews with no matching SentimentAnalysis for local-distilbert
2. Lazy-load DistilBERT model via singleton pattern, cached after first download
3. Run batch inference on review texts
4. Map model output to signed score: POSITIVE to +confidence, NEGATIVE to -confidence
5. Bulk insert results with skipDuplicates: true
6. Repeat until 0 pending reviews remain

P4.0 — Analytics and Reporting Engine

Purpose: Aggregates stored sentiment data into actionable intelligence metrics including time series, anomaly detection, and aspect-based analysis. **Trigger:** HTTP GET requests to analytics endpoints. **Logic:**

1. Overview: Count total/analyzed/pending reviews, calculate positive/negative percentages, aggregate rating distribution
2. Timeline: Group sentiments by time period using strftime, calculate per-period averages
3. Anomalies: Compute mean and standard deviation across timeline, flag data points where z-score exceeds threshold
4. Aspects: Keyword-match review text against 8 cinematic aspect categories, compute per-aspect sentiment scores

P5.0 — Frontend Presentation Layer

Purpose: Provides an interactive, real-time analytics dashboard rendered in the user browser. **Trigger:** User navigation to http://localhost:5174. **Logic:**

1. AuthContext manages login/register flow and stores token in localStorage
2. All API calls go through fetchWithAuth which injects Bearer token
3. Dashboard renders sentiment distribution, timeline charts, and anomaly alerts
4. Movie explorer provides search and pagination
5. Per-movie view shows detailed sentiment breakdown, aspect comparison, and review table with modal

End of DFD Document — SentixAI Platform v4.2 Enterprise